

Ke Wan

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PUBLICATIONS

- Cai, Z., Xiao, W., Sun, H., Luo, C., Zhang, Y., Wan, K., Li, Y., Zhou, Y., Chang, L.-W., Gu, J., Dong, Z., Anandkumar, A., Asi, A., Hu, J..
R-KV: Redundancy-aware KV Cache Compression for Reasoning Models.
39th Conference on Neural Information Processing Systems (NeurIPS 2025), Accepted, **peer-reviewed**.
<https://openreview.net/forum?id=2jwAjomEDB>
- Cai, Z., Qiu, H., Zhao, H., Wan, K., Li, J., Gu, J., Xiao, W., Peng, N., Hu, J..
From Preferences to Prejudice: The Role of Alignment Tuning in Shaping Social Bias in Video Diffusion Models.
Transactions on Machine Learning Research (TMLR), 2026, **peer-reviewed**.
<https://openreview.net/forum?id=C0yxuS6jty>
- He, Y., Chen, Q., Wan, K., Nabijiang, A., Cao, Y., Liu, B..
A Multi-Stage Machine Learning Pipeline for Automated Bowel Preparation Scale Assessment in Colonoscopy Videos.
2025 International Conference on Machine Learning and Applications (ICMLA 2025), 2026, **peer-reviewed**.
<https://ieeexplore.ieee.org/document/11471382>

PEER REVIEW SERVICE

- Neural Networks (Elsevier, Q1) — **15** completed manuscript reviews (2025 - 2026)
- IEEE Transactions on Image Processing (TIP) — **16** completed manuscript reviews (2025 - 2026)

EXPERIENCE

Software Engineer II

Microsoft

Oct. 2025 – Present

Sherman Oaks, CA

- Work on the design and development of **large-scale LLM inference platforms and AI agent serving systems**, supporting **AI intelligence** and **high-volume user-facing workloads** in production environments using **Python**, **Flask**, and **Kubernetes**.
- Support deployment of **flexible fine-tuned models with specialized knowledge**, addressing system complexity related to orchestration, performance stability, and scalability.
- Contribute to **SGLang-based LLM serving integration and inference orchestration**, focusing on scalability, reliability, and efficient runtime behavior under real-world operational constraints.
- Develop **LLM serving backends and internal APIs**, enabling robust serving orchestration and runtime management for internal AI platforms.

Software Engineer

Activision Blizzard Inc.

Jun. 2023 – Aug. 2025

Sherman Oaks, CA

- Contributed to Atlas, an anti-cheat **monorepo** for Game *Call of Duty: Black Ops 6* in 2024 with **Python** and **Golang**, processing **4.4M+** cheating events/hour at peak, **50%** improvement over 2023 platform.
- Built a **Kafka** consumer with **Golang** to fetch events with **Protobuf**, routing 25K messages/sec to Dead Letter Queues and API servers.
- Set up a **Flask** API server with 19 endpoints for querying events, deployed via **Gunicorn** on **Kubernetes** with **Horizontal Pod Autoscaling**, handling **20k** HTTP requests/sec.
- Developed a multi-threaded **AWS S3** Manager handling **6k** jobs/sec for persistent event storage, implemented metadata storage service for production use with **SQLAlchemy** across two databases.
- Refactored unit test modules using **Pytest**, achieving 100% coverage, enhanced **GitHub workflows** for pull-based **CI** unit testing and maintained **Docker** images for project's **CD** pipeline.

Software Engineer Intern

Jun. 2022 – Sept. 2022

Activision Blizzard Inc.

Santa Monica, CA

- Contributed to **Segmentation 2.0**, a real-time matchmaking system that grouped players by skill in predefined ratios with **Python**, improving matchmaking balance and game fairness over 1.0 version.
- Designed a calculation pipeline: produced user update events to **Kafka**, buffered via **RabbitMQ**, processed events and executed calculation with **Celery workers**, and stored results in **Redis**, handling **1.2M** events/day.
- Created 2 **RESTful APIs** in **Tornado** to get results from calculation pipeline, supporting **350k** requests/day.
- Self-driven a log monitoring and alerting system based on **Demonata** framework to gather, reformat, and transfer logs from matchmaking system to **Kibana** log platform, handing 17M logs per day, crafted **Grafana** dashboards to visualize log metrics and alerts.

Software Engineer Intern

Mar. 2021 – Jul. 2021

ByteDance Ltd.

Beijing, China

- Launched a reusable UI widget library with **Flutter** for ByteDance, migrating UI elements to a remote bucket, reducing app size by **20%**.
- Designed an app update SDK with ByteDance components to unify the codebase post-acquisition and enhance maintainability.
- Enforced multiple update services in **Flutter**, supporting multiple update modes including manual, automatic, and debug updates.
- Created 2 **gRPC APIs** using **Go** and **Protobuf** for fetching update package profiles and posting metrics/issues, integrated **Redis** to cache frequently used package metadata with 91% hit rate.

EDUCATION

University of California San Diego

Sep. 2021 – Mar. 2023

Master of Science in Computer Science | GPA: 3.97/4.0

La Jolla, CA

Beijing Jiaotong University

Sep. 2016 – Jul. 2020

Bachelor of Engineering in Computer Science | GPA: 3.79/4.0 | ranking: 1/219

Beijing, China

TECHNICAL SKILLS

Languages: Python, Go, Java, C/C++, SQL, Shell Script, Dart

Frameworks: Flask, Gunicorn, Pytest, PubSub, Kafka, Protobuf, Spark, Hadoop, Flutter, RabbitMQ, Tornado

Tools: Docker, Redis, GCP, Kubernetes, Git, GitHub Workflow, MySQL, Terraform, AWS